Using cosine formula, $\cos C = \dfrac{a^2+b^2-c^2}{2ab}$

$$\cos 75^\circ = \frac{8+12-c^2}{8\sqrt{6}} \Rightarrow \frac{\sqrt{3}-1}{2\sqrt{2}} = \frac{8+12-c^2}{8\sqrt{6}} \Rightarrow c = \sqrt{2}\left(\sqrt{3}+1\right)$$

Also, $\cos A = \dfrac{b^2+c^2-a^2}{2bc} = \dfrac{1}{\sqrt{2}}$, Thus $A = 60^\circ,\ B = 180^\circ - (A+C) = 45^\circ$.

Example 3.67 Find the area of the triangle whose sides are $13\ cm, 14\ cm$ and $15\ cm$.

Solution:

We know that $s = \dfrac{a+b+c}{2} \Rightarrow s = \dfrac{13+14+15}{2} = 21\ cm.$

Area of a triangle $= \sqrt{s(s-a)(s-b)(s-c)}$

$= \sqrt{21(21-13)(21-14)(21-15)} = 84$ sq.cm.

Example 3.68 In any $\triangle ABC$, prove that $a\cos A + b\cos B + c\cos C = \dfrac{8\triangle^2}{abc}$.

Solution:

We know that $a\cos A + b\cos B + c\cos C = 2a\sin B\sin C$

Thus, $a\cos A + b\cos B + c\cos C = 2a\left(\dfrac{2\triangle}{ac}\right)\left(\dfrac{2\triangle}{ab}\right) = \dfrac{8\triangle^2}{abc}$

Example 3.69

Suppose that there are two cell phone towers within range of a cell phone. The two towers are located at $6\ km$ apart along a straight highway, running east to west and the cell phone is north of the highway. The signal is $5\ km$ from the first tower and $\sqrt{31}\ km$ from the second tower. Determine the position of the cell phone north and east of the first tower and how far it is from the highway.

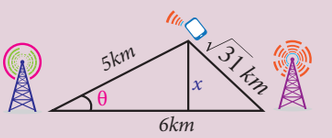


Figure 3.24

Solution:

Let θ be the position of the cell phone from north to east of the first tower.

Then, using the cosine formula, we have,

$$\begin{aligned}\left(\sqrt{31}\right)^2 &= 5^2+6^2-2\times5\times6\cos\theta\\ 31 &= 25+36-60\cos\theta\\ \cos\theta &= \frac{1}{2} \Rightarrow \theta = 60^\circ\end{aligned}$$

Let x be the distance of the cell phone's position from the highway.

Then, $\sin\theta = \dfrac{x}{5} \Rightarrow x = 5\sin\theta = 5\sin 60^\circ = \dfrac{5\times\sqrt{3}}{2}\ km.$